Learning Compositional Tasks via Trigger Compositions: Using Scratchpads as Pre-Answer Workspaces

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Abstract

Trigger tokens promise reusable, composable control: a model should apply learned operations to new inputs, extend them to longer trigger sequences, and sometimes answer without showing intermediate work. We study these expectations in a fully verifiable arithmetic trigger-following task. We call visible intermediate text generated before the final answer a *scratchpad*; direct answering asks for the final answer alone, without the scratchpad. Across multi-seed fine-tuning experiments, pre-answer scratchpads enable reliable composed execution on held-out inputs, while direct-answer and answer-first formats remain weak. A content-matched placement ablation shows that the same symbolic work is far less effective after the answer than before it, indicating that scratchpads help as pre-answer workspaces rather than as reasoning-like text appended anywhere. However, this is not a simple "scratchpads always generalize" story: when training excludes all three-trigger sequences, verbose self-contained step descriptions generalize better than the tested compact symbolic scratchpad. Finally, mixed-format training and a no-finetuning prompting baseline do not yield reliable direct answering without the scratchpad. These findings suggest that compositional trigger following depends on the output interface for intermediate state: where the scratchpad is placed, how its steps are structured, and whether explicit work can be converted into answer-only behavior.

1. Introduction

Instruction-tuned language models are increasingly used as controllable systems. In practical interfaces, long instructions are often compressed into short tags, control codes, or trigger tokens that specify a desired domain, audience, style, safety policy, or output format. A sequence such as `[FINCORP][EXTERNAL][SUMMARY]` might abbreviate a corporate-finance framing for an external audience in a concise style; `[SUPPORT][FRIENDLY][JSON]` might request a friendly customer-support response in a structured schema. Such interfaces are useful only to the extent that the controls are reusable and composable.

We use *scratchpad* in a concrete way. A scratchpad is ordinary visible output text, generated before the final answer, that records intermediate state. It is not a special hidden reasoning token and it is not merely any explanation that appears somewhere in the output. For example, if $[A][B][C]$ applied to 51 gives $51 \to 52 \to 50 \to 47$, a scratchpad may write the intermediate values 52 and 50 before returning 47. This definition is deliberately operational: a scratchpad is a workspace the model can condition on before producing the answer. By contrast, *direct answering* asks the model to output only the final answer, with no scratchpad. Scratchpads are also practically attractive because they are human-readable, debuggable, and learnable through standard fine-tuning.

This paper studies three transfer assumptions implicit in trigger-based control. The first is *execution under input shift*: after seeing a trigger composition on one set of inputs, can the model execute it on new inputs? The second is *length generalization*: after seeing one- and two-trigger compositions, can the model extend the learned procedure to held-out three-trigger compositions? The third is *direct answering without the scratchpad*: if scratchpad supervision teaches the model how to compose triggers, does the same skill appear when the model is asked to output only the final answer?

We study these questions in a controlled arithmetic proxy. Natural-language trigger following is hard to score exactly because style, audience, and domain adherence are partly subjective. Arithmetic triggers give us a small but fully verifiable setting: each trigger is a deterministic state update on an integer, each ordered trigger sequence has a unique correct output, and output formats can be compared while holding the underlying task fixed. We instantiate three trig-

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gers with distinct predicates: parity, digit-sum divisibility by 3, and divisibility by 5.

The central design variable is the output interface for intermediate state. Our experiments ask when a scratchpad functions as a workspace that the model can use before committing to the final answer. This suggests a distinction that is easy to miss in broad discussions of "reasoning": a pre-answer scratchpad, a verbose explanation, a compact symbolic marker, and a post-answer verification are not interchangeable. They differ in what information is available before the answer token is generated and in whether the scratchpad exposes a step structure that can be extended to longer compositions.

Our results support this more structured view. On held-out inputs, both compact symbolic `brief` scratchpads and verbose natural-language `full` scratchpads reach near-ceiling accuracy at depth 3, whereas direct-answer and answer-first `verify` formats remain weak and seed-sensitive. A content-matched placement ablation then isolates ordering: the same symbolic work that solves the task when generated before the answer collapses when moved after the answer. However, a held-out-depth experiment qualifies the story: when all depth-3 compositions are removed from training, `full` generalizes to depth 3 but `brief` largely does not. Pre-answer placement is therefore the most reliable regime among the tested formats in this probe, but it is not sufficient for length generalization; the tested scratchpad also appears to need an extensible step representation.

Our contributions are:

1. **A transfer framing for compositional trigger following.** We distinguish execution under input shift, length generalization to held-out composition depth, and transfer from scratchpad-based execution to direct answering without the scratchpad.

2. **A fully verifiable trigger-following probe with distinct predicates.** We use deterministic arithmetic triggers over parity, digit-sum divisibility by 3, and divisibility by 5, allowing exact evaluation of both answer accuracy and output-format effects.

3. **Evidence that pre-answer workspaces support execution but do not automatically yield direct answering without the scratchpad.** In three-seed replicated conditions, `brief` and `full` pre-answer channels are stable and near-perfect on held-out inputs, while the direct channel remains weak even in mixed-format models.

4. **A content-matched placement ablation.** Moving the same symbolic work from before the answer to after the answer removes most of its benefit, separating a pre-answer workspace from post-hoc reasoning-like text.

5. **A length-generalization result that refines the compact-scratchpad story.** The tested compact symbolic scratchpad performs well for seen composition depths on held-out inputs, but verbose self-contained step descriptions are much more robust when depth-3 compositions are held out.

6. **A no-finetuning prompting baseline.** Five-shot prompting on the same base model fails to reproduce the fine-tuned scratchpad gains, suggesting that the effect is not explained by in-context exposure to the target format alone.

The claim is intentionally narrow. We do not argue that scratchpads solve compositional generalization in general, nor that arithmetic captures the full complexity of natural-language control. We show that, in a controlled trigger-following probe, composed execution is strongest when intermediate state is generated before answer commitment, and held-out-depth generalization further depends on whether the scratchpad exposes a step structure that can be extended.

2. Related Work

Compositional generalization. Compositional generalization benchmarks such as SCAN, COGS, and CFQ show that neural models can master familiar pieces while failing on systematic recombinations (Lake & Baroni, 2018; Kim & Linzen, 2020; Keysers et al., 2020). Our probe targets an analogous failure mode but trades ecological realism for control: the operations are deterministic arithmetic triggers rather than semantic parses or action sequences, so every output can be checked exactly and the supervision format can be varied without changing the task.

Scratchpads, chain-of-thought, and process supervision. Scratchpads expose intermediate computation as tokens (Nye et al., 2021); chain-of-thought prompting shows that intermediate reasoning can improve arithmetic, symbolic, and logical tasks (Wei et al., 2022; Kojima et al., 2022); and STaR-like methods learn from generated rationales (Zelikman et al., 2022). Process-supervision work further shows that step-level feedback can outperform final-answer-only supervision on math tasks (Lightman et al., 2024). Our study is not a first claim that reasoning helps. It asks which token-level interface makes intermediate state useful: answer-only, compact pre-answer, verbose pre-answer, or post-answer verification.

Composing learned skills with scratchpads. Recent work is especially close to our motivation. Yao et al. (2025)

study compositional generalization from learned skills via CoT training, and Yin et al. (2025) show that simply training on atomic CoT is limited unless the scratchpad format itself is composable. Our contribution is narrower and diagnostic: we compare fixed output channels in a controlled trigger-following probe, add a content-matched pre- versus post-answer placement ablation, and separate held-out-input execution from held-out-depth generalization.

Compact scratchpads and internalization. Chain of Draft argues that concise intermediate drafts can match verbose CoT while using far fewer tokens (Xu et al., 2025). Our held-out-input results are consistent with that view: compact `brief` and verbose `full` both reach near-ceiling accuracy. The held-out-depth results qualify it: compactness alone does not guarantee extension beyond trained composition depths. A separate line of work asks whether explicit reasoning can be internalized into direct answering, for example through stepwise removal of CoT or latent/quiet reasoning mechanisms (Deng et al., 2024; Zelikman et al., 2024). Our negative direct-channel results suggest that direct answering without the scratchpad is not an automatic consequence of mixing explicit and direct supervision.

Arithmetic as a diagnostic proxy. A recent meta-analysis finds that CoT benefits are strongest on math and symbolic reasoning tasks (Sprague et al., 2024). We deliberately use such a setting because it permits exact evaluation and clean diagnosis of format effects. The intended application is compositional control, so external validity to natural-language trigger systems remains an empirical question rather than an assumption.

3. A Verifiable Trigger-Following Task

3.1. Trigger rules

We define three deterministic trigger functions over integer inputs. A composition applies them left-to-right.

- **A:** if n is odd, set $n \leftarrow n+1$; otherwise set $n \leftarrow n+2$.
- **B:** if the digit sum of n is divisible by 3, set $n \leftarrow 2n$; otherwise set $n \leftarrow n-2$.
- **C:** if n is divisible by 5, set $n \leftarrow n-3$; otherwise set $n \leftarrow n+10$.

For example, $[A][B][C]$ applied to 51 yields $51 \rightarrow 52 \rightarrow 50 \rightarrow 47$. The task is fully verifiable because every trigger sequence and input has a unique correct output. The three predicates are deliberately distinct: A depends on parity, B on digit-sum divisibility by 3, and C on divisibility by 5, so success cannot be explained by two triggers sharing the same branching condition. Composition depth and trigger order can be varied independently of input number, which we use in Section 4 to define complementary training and evaluation splits.

Table 1. Output formats. A scratchpad is visible intermediate text written before the final answer. The bottom row is introduced for the placement ablation: it uses the same symbolic intermediate work as `brief` but places it after the answer.

Format	Example for [A] 51	Role in the ablation
`direct`	`52`	final answer only
`brief`	`odd+1=52`	compact symbolic state, pre-answer
`short`	`o 52`	maximally compressed state, pre-answer
`full`	`51 is odd, so add 1: 51+1=52. Answer: 52.`	verbose natural-language state, pre-answer
`verify`	`Answer: 52. Check: [A]: if odd +1, else +2. ...`	answer-first verification
`bp`	`Answer: 52. Reasoning: odd+1=52`	same symbolic work as `brief`, post-answer

3.2. Output formats

We fine-tune the same base model with six target formats. These formats differ in two conceptually separate ways: whether they expose intermediate state, and whether that state appears before or after the final answer. In this paper, only pre-answer intermediate text is called a scratchpad; post-answer text may contain reasoning-like content, but it is not a workspace for producing the already-emitted answer. The first five formats form the base ablation grid; the sixth, `brief-post`, is introduced specifically to isolate placement.

For compositions, each format extends naturally. For example, `brief` for $[A][B][C]$ applied to 51 becomes `odd+1=52, ds¬3−2=50, div5−3=47`, followed by the final answer. The corresponding `brief-post` target is `Answer: 47. Reasoning: odd+1=52, ds¬3−2=50, div5−3=47`. At evaluation time, `brief-post` accuracy is computed from the leading answer field only. The post-answer reasoning text therefore cannot repair an answer that has already been emitted.

3.3. Training mixtures

We study nine mixtures of output formats. The labels d, b, s, f, v, bp denote `direct`, `brief`, `short`, `full`, `verify`, and `brief-post`, respectively.

E1: d **E2:** d+b **E3:** d+s **E4:** d+f **E5:** d+v **E6:** d+f+v **E7:** d+b+f **E8:** d+b+v **E9:** d+bp.

E9 is the placement ablation. It matches E2 in its trained formats except that the pre-answer `brief` channel is replaced by the post-answer `brief-post` channel.

3.4. Metrics

Per-format accuracy. If a model was trained on a format f, we evaluate it by requesting that format at test time and report $\mathrm{Acc}(f, k)$ for composition depth k.

Aggregate composition accuracy. Let F be the set of formats used during training for a given experiment. We define

$$\mathrm{AggAcc@}k = \frac{1}{|F|} \sum_{f \in F} \mathrm{Acc}(f, k).$$

This metric summarizes how well a trained model performs across all output channels it was taught to use. It is an average over trained formats, not a best-of score. This distinction matters because aggregate accuracy can be high even when the direct channel is weak.

3.5. Implementation details

Base model. LLaMA-3.2-3B-Instruct (4-bit).

Fine-tuning. LoRA via Unsloth, effective batch size 16, learning rate 5×10^{-5}.

Training schedule. All multi-seed experiments use 40 epochs with early stopping on `eval_loss` (patience 3); models therefore stop at different points across seeds based on validation behavior.

Data. The dataset contains 15 trigger compositions (3 atomic, 6 ordered length-2, and 6 ordered length-3). Our main experiments share these compositions between training and evaluation, splitting only by input number (1–50 for training and 51–70 for evaluation). Section 4 therefore measures generalization to held-out inputs at fixed composition depth. To complement this with a held-out-composition test, we also run a length-generalization sweep where training compositions are restricted to depths 1 and 2 (9 compositions) and depth-3 compositions are evaluated as a held-out set; results are reported in Section 4.4.

Seeds. E1–E5, E7, and the placement ablation E9 are run with three seeds $\{0, 1, 42\}$; all reported numbers for these experiments are mean $\pm$ standard deviation across seeds. E6 and E8 are reported with seed 42 only and are included for completeness rather than as load-bearing evidence. Thus E9 is a replicated load-bearing ablation, not a seed-42-only result.

4. Results

4.1. Pre-answer scratchpads enable composed execution under input shift

Table 2 isolates the main comparison: direct-only supervision versus direct plus one non-direct output format. All numbers are averaged over three seeds.

The result is not a simple length effect. Direct-answer supervision collapses with depth-3 compositions and is seed-sensitive ($19.7 \pm 11.4\%$). By contrast, both pre-answer scratchpad formats reach near-ceiling accuracy: `brief` reaches $98.6 \pm 2.4\%$ and `full` reaches $99.4 \pm 0.5\%$. These formats differ substantially in verbosity and surface form, but they share an operational property: intermediate state is generated before the final answer. The answer-first `verify` channel lacks that property and performs near the direct baseline ($25.8 \pm 10.1\%$).

This establishes the positive result for the held-out-input setting: the model executes composed triggers reliably on new inputs when allowed to compute explicitly before answering. Whether this skill extends to held-out compositions is the question of Section 4.4.

4.2. Aggregate gains do not imply direct answering without the scratchpad

Table 3 reports the strongest replicated mixture, E7 = d+b+f. Table 4 reports two additional single-seed mixtures for completeness.

E7 achieves the best aggregate depth-3 score, but the channel breakdown changes the interpretation. The gain comes from scratchpad channels: `brief` and `full` remain near-perfect. The direct channel of the same trained model is weak ($9.7 \pm 6.9\%$). Thus, scratchpad supervision does teach a compositional skill, but the skill is expressed through the channels that allow pre-answer computation. It does not reliably become a direct-answer capability without the scratchpad under this training recipe.

4.3. Content-matched placement ablation

The comparisons above suggest that pre-answer placement matters, but `brief`, `full`, and `verify` differ along several axes at once: compactness, naturalness, verbosity, and ordering. To isolate ordering, we match the symbolic scratchpad content and vary placement. The `brief` format emits a symbolic scratchpad before the answer, for example `odd+1=52, ds¬3−2=50, div5−3=47`. The `brief-post` format emits the same symbolic work after the answer: `Answer: 47. Reasoning: odd+1=52, ds¬3−2=50, div5−3=47`.

Table 5 shows the result. With the symbolic scratchpad con-

Table 2. Single-format comparison on 3-trigger composition. Mean ± stdev over n=3 seeds. All values are percent.

Exp	Train formats	AggAcc@3	d@3	Explicit channel @3
E1	`d`	19.7 ± 11.4	19.7 ± 11.4	—
E2	`d+b`	60.6 ± 7.6	22.5 ± 16.0	**b:** 98.6 ± 2.4
E3	`d+s`	56.8 ± 4.8	21.1 ± 11.1	s: 92.5 ± 3.6
E4	`d+f`	62.9 ± 4.8	26.4 ± 9.3	**f:** 99.4 ± 0.5
E5	`d+v`	27.1 ± 8.7	28.3 ± 7.3	v: 25.8 ± 10.1

Table 3. E7 mixture on 3-trigger composition. Mean ± stdev over n=3 seeds.

Exp	Train formats	AggAcc@3	d@3	Other channels @3
E7	`d+b+f`	$\mathbf{68.9 \pm 2.3}$	9.7 ± 6.9	b: 98.9 ± 0.5, f: 98.1 ± 1.0

Table 4. Additional mixtures, single-seed (seed 42). Multi-seed replication for these two auxiliary mixtures remains future work.

Exp	Fmt	Agg@3	d@3	Other @3
E6	`d+f+v`	41.7	15.0	f: 95.0, v: 15.0
E8	`d+b+v`	35.0	2.5	b: 99.2, v: 3.3

tent matched, moving the same symbolic work from before the answer to after the answer reduces depth-3 accuracy by approximately 70 percentage points.

The content-matched comparison supports a placement-based explanation. The matched symbolic work that nearly solves the composition when emitted before the answer becomes largely inert when emitted after it. Post-answer reasoning text, whether compact and symbolic (`brief-post`) or verbose and natural-language (`verify`), falls near direct-answer performance. This does not show that every future trigger-following task will behave the same way, nor does it remove the need for within-checkpoint and cross-model probes. It does show that in this controlled probe, the useful scratchpad property is not mere target-side reasoning text. The useful property is pre-answer intermediate state that the model can condition on before answer commitment.

The collapse is also depth-dependent. In seed 42, `brief-post` achieves 81.7% at depth 1 and 63.3% at depth 2, before collapsing to the depth-3 levels reported in Table 5. This pattern is consistent with an early-commitment failure: once the model must answer before explicitly updating the intermediate state, errors compound as the number of triggers grows.

4.4. Length generalization: depth-1+2 train, depth-3 held-out

The experiments above test how reasoning format affects composed execution when the same trigger compositions appear in train and eval. We now ask the held-out-depth question. We retrain E1, E2, E4, E5, and E7 with a modified split: training compositions are restricted to depths 1 and 2 (the 3 atomic triggers and the 6 ordered length-2 sequences); the 6 ordered length-3 compositions never appear during training and are evaluated as a held-out set. Inputs span 1–70 in both splits, isolating the composition-depth axis. All other hyperparameters match the main sweep, with seeds $\{0, 1, 42\}$.

The two pre-answer formats, which were nearly indistinguishable on held-out inputs (`brief`: 98.6 ± 2.4, `full`: 99.4 ± 0.5), now diverge by more than 60 percentage points. Verbose natural-language reasoning (`full`) generalizes to held-out depth-3 compositions at $83.8 \pm 10.4\%$ in E4 and $93.1 \pm 1.2\%$ in E7. Compact symbolic reasoning (`brief`) collapses to $23.2 \pm 2.5\%$ in E2, recovering only partially to $41.1 \pm 11.9\%$ when co-trained with `full` in E7. The answer-only and answer-first channels (`direct`, `verify`) drop near 10%, near the floor for exact-answer performance on the held-out triples.

This sharpens the picture from the held-out-input experiments. Among the tested formats, pre-answer placement is the most reliable regime for composed execution, but it is not sufficient for length generalization: the scratchpad also appears to need a step representation the model can extend. Verbose self-contained step descriptions (e.g., “51 is odd, so add 1: $51 + 1 = 52$.”) support that extension; the tested compact symbolic markers do not. Mixing verbose and compact training partially carries the length-generalization skill into the compact channel, but does not close the gap. The placement claim from Table 5 is unaffected by this distinction — it is a within-format comparison — but the format ranking changes once held-out depth is part of the evaluation.

Pre-answer scratchpad channels are not just better on average; they are also more reliable across seeds. Table 7 ranks the most stable load-bearing channels and the most variable ones.

The low variance of the pre-answer channels strengthens the interpretation of the positive result. The high variance of

Table 5. Placement ablation. The first two rows match symbolic scratchpad content and vary placement. `brief-post` collapses toward direct-answer and answer-first verification performance. All rows are mean ± stdev across three seeds $\{0, 1, 42\}$.

Variant	Content	Placement	depth-3 accuracy
`brief` (E2)	symbolic	pre-answer	$\mathbf{98.6 \pm 2.4}$
`brief-post` (E9)	**symbolic (matched)**	**post-answer**	$\mathbf{32.2 \pm 3.9}$
`verify` (E5)	verbose	post-answer	25.8 ± 10.1
`direct` (E9, same model)	none	—	29.2 ± 2.5
`direct` (E1)	none	—	19.7 ± 11.4

Table 6. Length generalization. Training compositions are depth 1+2 only; depth-3 compositions are held out. Mean ± stdev across n=3 seeds. The two pre-answer formats diverge sharply: `full` preserves most of its accuracy, while `brief` collapses.

Exp	Train fmt	d@3	b@3	f@3	v@3
E1	`d`	9.4 ± 3.2	—	—	—
E2	`d+b`	9.1 ± 1.8	23.2 ± 2.5	—	—
E4	`d+f`	3.4 ± 1.6	—	$\mathbf{83.8} \pm \mathbf{10.4}$	—
E5	`d+v`	8.9 ± 0.8	—	—	10.2 ± 2.1
E7	`d+b+f`	4.5 ± 1.6	41.1 ± 11.9	$\mathbf{93.1} \pm \mathbf{1.2}$	—

Table 7. Stability ranking across the three seeds. All values are percent. Pre-answer load-bearing channels are tightly clustered; direct and verification channels are much noisier.

Rank	Exp	Channel	Mean	Stdev	Range
1	E4	`full@3`	99.4	0.5	0.8
2	E7	`brief@3`	98.9	0.5	0.8
3	E7	`full@3`	98.1	1.0	1.7
4	E2	`brief@3`	98.6	2.4	4.2
5	E3	`short@3`	92.5	3.6	6.7
6	E7	`direct@3`	9.7	6.9	13.3
7	E5	`verify@3`	25.8	10.1	17.5
8	E1	`direct@3`	19.7	11.4	20.8
9	E2	`direct@3`	22.5	16.0	29.2

direct and verification channels also cautions against over-interpreting small differences among weak channels. The robust distinction is between pre-answer computation and answer-first or answer-only generation.

4.5. Prompting baseline: in-context format exposure is not enough

The preceding results compare fine-tuned output formats. We additionally test whether the strongest scratchpad results could be explained by in-context exposure to the desired format without fine-tuning. Using the same base model without LoRA, we run greedy five-shot prompting on a 60-example depth-3 subset (the six ordered triples over inputs 51–60). The five demonstrations are format-specific and include correct depth-1, depth-2, and depth-3 worked examples, so the prompt exposes both the arithmetic rules and the requested output style.

Table 8 shows that prompting alone does not approach the fine-tuned scratchpad channels. The prompted `brief` format scores 0.0% despite format-matched demonstrations; verbose prompting reaches only 11.7%, far below the fine-tuned `brief` and `full` channels. Qualitatively, the base model often copies surface labels from the demonstrations rather than re-evaluating the trigger predicates on the new input. This baseline is intentionally modest: one small model, greedy decoding, one five-shot template, and a 60-example subset. Still, it addresses the most direct alternative explanation for the main result: in this setting, the fine-tuned gains are not reproduced by in-context format exposure alone.

Format	5-shot prompt	Fine-tuned	Gap
`direct`	0.0	19.7 ± 11.4	-19.7
`brief`	0.0	98.6 ± 2.4	-98.6
`full`	11.7	99.4 ± 0.5	-87.7

Table 8. No-finetuning prompting baseline on a 60-example depth-3 subset. Prompting uses the same base model, greedy decoding, and five correct in-context demonstrations. Fine-tuned numbers are the corresponding three-seed depth-3 channels from the main sweep. All values are percent.

5. Analysis

5.1. What transfers, and through which channel?

The experiments separate three kinds of transfer. The first is *composed execution under input shift*: a model trained on the full set of compositions across one numeric range applies them on a held-out numeric range. The second is *length generalization*: a model trained on compositions of depth 1 and 2 applies them to held-out depth-3 compositions. The third is *direct answering without the scratchpad*: a model that has learned to compose through an explicit scratchpad later expresses the same skill by outputting only the final answer.

Our positive result concerns the first kind of transfer. The `brief` and `full` channels both reach near-ceiling on three-trigger compositions evaluated on held-out inputs, showing that pre-answer scratchpads support reliable composed execution under input shift. Our second result is a partial answer to length generalization: `full` extends to held-out depth-3 compositions at 83–93% while `brief` collapses, indicating that pre-answer placement alone is not sufficient in this tested setting and that self-contained step descriptions provide an additional advantage. Our negative result concerns the third kind. E7 solves the task through its explicit channels but remains weak in its direct channel. The learned skill is therefore channel-dependent: it appears when the inference format permits pre-answer computation, but it does not reliably appear when the model is asked to answer without the scratchpad.

5.2. Why pre-answer placement matters

Autoregressive generation makes placement operationally important. In a pre-answer scratchpad, each generated intermediate value becomes part of the context for the next step and for the final answer. For $[A][B][C]$ applied to 51, the model can generate 52, then condition on 52 to generate 50, then condition on 50 to generate 47. The answer is produced after the intermediate state has been made available in the context.

In a post-answer format, the ordering is reversed. The model is asked to emit `Answer: 47` before it has generated the intermediate work that would produce 47. Any subsequent reasoning may be locally consistent or fluent, but it cannot change the already-committed answer under standard left-to-right decoding. This explains why `brief-post` and `verify` behave like answer-first channels rather than like scratchpad channels. They include reasoning-like text, but the reasoning comes too late to serve as computation for the answer.

This mechanism remains a hypothesis that should be tested directly by decoding and auditing failed post-answer reasoning. For example, one can check whether the post-answer work computes a different value from the leading answer, or whether it rationalizes the wrong answer. The current results nevertheless identify a strong design constraint: target formats should be evaluated by what information is available before the answer, not only by whether the final target contains reasoning text somewhere.

5.3. Atomic mastery does not guarantee composition

Direct-only supervision can teach atomic and short-composition cases, but it does not force reliable recombination at depth 2 or 3 even when those compositions are seen during training. In the direct format, the model never emits the intermediate state that should become the next trigger's input. At inference time it has to perform the entire update sequence internally, even though the supervised target does not expose that computation. The resulting failure mirrors broader compositional generalization results: local competence on familiar pieces does not guarantee systematic recombination of those pieces on new inputs or at longer depths.

5.4. Extensible step structure matters for held-out depth

The two pre-answer formats, `brief` and `full`, behave similarly on held-out inputs (98.6 vs 99.4%) despite a large difference in verbosity. This might suggest that pre-answer placement alone is the key variable. The length-generalization experiments in Section 4.4 qualify that conclusion. On held-out depth-3 compositions, the two formats diverge by more than 60 percentage points: `full` retains 83–93% while `brief` drops to 23–41%.

We interpret this as evidence about *extensible step structure*, not raw word count. The successful `full` format expresses each trigger application as a self-contained subproblem ("51 is odd, so add 1: $51 + 1 = 52$."), which can be repeated one more time at depth 3. The compact `brief` markers expose intermediate values but may fit the depth-1/depth-2 surface patterns without making the reusable step schema explicit. A better symbolic format might close this gap; our data only show that the particular compact scratchpad tested here does not. The `short` format, which lags `brief` and `full` even on held-out inputs, is consistent with the same interpretation: its one-letter codes remove predicate identifiers and push the model toward pattern matching rather than reusable step execution.

5.5. No reliable transfer to direct answering without the scratchpad

The mixture experiments show that scratchpad supervision can create a high-performing scratchpad channel without improving answer-only generation. E7 achieves $68.9 \pm 2.3\%$ aggregate depth-3 accuracy, but this number averages over one weak direct channel and two near-perfect explicit channels. Its direct channel alone is $9.7 \pm 6.9\%$. Thus, the aggregate score should not be read as evidence that the model has internalized a general compositional ability that works without the scratchpad.

This matters for deployment. A system designer might train with scratchpads and hope to remove them at inference time while preserving the learned compositional behavior. Our results do not support that assumption in this setting. If answer-only deployment is desired, it should be treated as a separate distillation problem rather than an automatic consequence of scratchpad training.

5.6. A unified picture: explicit workspace, not automatic direct answering

The results support a layered partition of the supervision space. On held-out inputs, pre-answer scratchpad channels reach 92–99% depth-3 accuracy, answer-first reasoning channels reach roughly 26–32%, and answer-only channels are weak and noisy: placement separates the high-performing and low-performing regimes regardless of whether the scratchpad is compact or verbose. On held-out compositions, the partition refines further within the pre-answer cluster: verbose pre-answer scratchpads preserve 83–93% while compact pre-answer scratchpads drop to 23–41%. Across the tested formats, pre-answer scratchpad channels are the consistently high-performing channels under input shift; when the evaluation depth exceeds the training depth, the tested verbose self-contained step descriptions provide an additional advantage over the tested compact scratchpad.

This is the sense in which successful trigger transfer remains tied to an explicit scratchpad under the training recipe tested here. The model can compose learned trigger behaviors when the output format gives it an explicit computation channel before the answer. The same learned behavior does not reliably appear when that channel is absent or when the intermediate work is moved after answer commitment.

6. Limitations

Several limitations should be stated plainly.

Single model family. All experiments use LLaMA-3.2-3B-Instruct (4-bit). We do not yet know whether the same placement effect or format ranking holds for other model families, scales, or decoding settings.

Composition depth bounded at three. We evaluate at depths 1, 2, and 3. Deployed trigger systems may stack four, five, or more controls. Whether pre-answer scratchpads continue to generalize at greater depth, and whether the post-answer collapse becomes catastrophic earlier, remains open.

Prompting baseline scope. Section 4.5 rules out the simplest no-finetuning alternative on the same small base model: five-shot prompting does not reproduce the fine-tuned scratchpad gains. However, this is not an exhaustive prompting study. Larger models, different templates, self-consistency, tool use, or more demonstrations may narrow the gap.

Placement is content-matched but not within-checkpoint. The placement comparison contrasts the `brief` channel of the E2-trained model with the `brief-post` channel of a separately trained E9 model using matched symbolic content and the same seeds. Within E9 itself, the direct channel ($29.2 \pm 2.5\%$) and the trailing `brief-post` channel ($32.2 \pm 3.9\%$) are close, which is consistent with our placement interpretation. A stricter within-checkpoint probe that prompts the same model with both pre- and post-answer formats would further tighten the placement claim.

Additional mixtures still single-seed. The additional mixtures E6 and E8 use a single seed (42). They are reported for completeness in Appendix A but are not load-bearing for any claim in the main text.

Proxy task. Arithmetic provides exact evaluation, but the intended application is compositional control of natural-language behavior. A natural-language pilot with decomposed domain, audience, style, and format triggers would be needed to test external validity.

Pending failure-mode analysis. The mechanism proposed in Section 5.2 should be tested by inspecting failed `brief-post` and `verify` generations. In particular, future work should check whether post-answer reasoning computes the correct value after an incorrect leading answer, or instead rationalizes the wrong answer.

7. Conclusion

This paper tested three transfer assumptions behind trigger-based control: execution under input shift, length generalization to held-out composition depth, and direct answering without the scratchpad. The answers are not the same. In our controlled arithmetic probe, pre-answer scratchpads are the consistently high-performing channels on held-out inputs: compact symbolic and verbose natural-language scratchpads both solve depth-3 compositions nearly perfectly. But when all depth-3 trigger sequences are held out during training, the formats diverge: verbose self-contained step descriptions generalize, while the compact symbolic scratchpad largely collapses.

The placement ablation clarifies why answer-first reasoning text is a poor substitute for a scratchpad. Holding symbolic scratchpad content fixed, moving the same symbolic work from before the answer to after the answer removes most of its benefit and brings performance close to direct answering. Thus the useful role of a scratchpad in this setting is not merely to add reasoning-like text to the target. It is to make the model generate and condition on intermediate state before committing to the final answer.

The practical implication is that explicit scratchpads should not be treated as disposable training scaffolds for compositional trigger-following systems. Under the training recipe tested here, strong explicit channels do not compile themselves into reliable answer-only inference. If a deployment calls for direct answers, it should include an explicit internalization or distillation stage; otherwise, the operational interface should preserve a pre-answer computation channel. More broadly, the results suggest that scratchpad design should be evaluated along at least three axes: placement before the answer, extensibility of the step representation, and transfer into direct answering without the scratchpad.

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A. Three-trigger results, multi-seed and single-seed

This appendix provides the per-channel breakdown underlying Tables 2, 3, 5 and 7. Rows marked n=3 are reported as mean $\pm$ stdev over seeds $\{0, 1, 42\}$. Rows marked seed 42 are single-seed auxiliary mixtures included for completeness.

Exp	Train fmt	Agg@3	d@3	b@3	s@3	f@3	v@3	Notes
E1	`d`	19.7 ± 11.4	19.7 ± 11.4	—	—	—	—	n=3
E2	`d+b`	60.6 ± 7.6	22.5 ± 16.0	98.6 ± 2.4	—	—	—	n=3
E3	`d+s`	56.8 ± 4.8	21.1 ± 11.1	—	92.5 ± 3.6	—	—	n=3
E4	`d+f`	62.9 ± 4.8	26.4 ± 9.3	—	—	99.4 ± 0.5	—	n=3
E5	`d+v`	27.1 ± 8.7	28.3 ± 7.3	—	—	—	25.8 ± 10.1	n=3
E7	`d+b+f`	68.9 ± 2.3	9.7 ± 6.9	98.9 ± 0.5	—	98.1 ± 1.0	—	n=3
E9	**`d+bp`**	$\mathbf{30.7 \pm 2.1}$	$\mathbf{29.2 \pm 2.5}$	— (bp: $\mathbf{32.2 \pm 3.9}$)	—	—	—	**n=3**
E6	`d+f+v`	41.7	15.0	—	—	95.0	15.0	seed 42
E8	`d+b+v`	35.0	2.5	99.2	—	—	3.3	seed 42

Table 9. Full three-trigger results. Rows marked n=3 are mean $\pm$ stdev across seeds $\{0, 1, 42\}$; rows marked seed 42 are single-seed auxiliary mixtures. For E9, the `brief-post` (bp) channel score is $32.2 \pm 3.9\%$, included in AggAcc@3 with d@3.